



PowerPoint Presentation by
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MANAGEMENT ACCOUNTING

8th EDITION

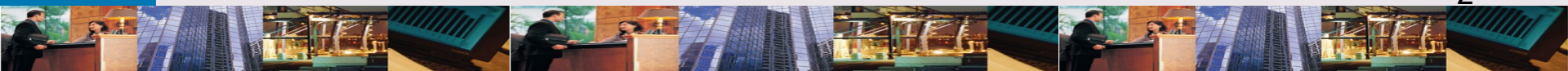
BY

HANSEN & MOWEN

5 ACTIVITY-BASED MANAGEMENT

LEARNING OBJECTIVES

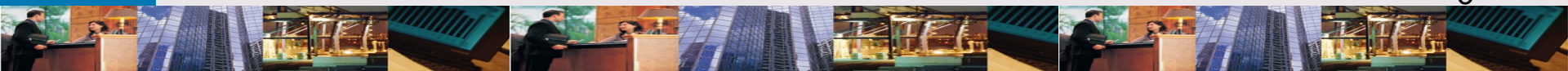
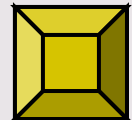
After studying this chapter, you should be able to:



LEARNING OBJECTIVES

1. Describe activity-based management & explain its relationship to activity-based costing.
2. Explain process value analysis.
3. Describe activity performance measurement.
4. Describe activity-based customer & supplier costing

*Click the button to skip
Questions to Think About*



QUESTIONS TO THINK ABOUT: Roberts Truck Products (RTP)

Why is accurate cost information
about customers & suppliers
important?

What is wrong with the claim that
all customers are good customers?



QUESTIONS TO THINK ABOUT: Roberts Truck Products (RTP)

Will accurate cost information
guarantee that a firm is
competitive?



QUESTIONS TO THINK ABOUT:

Roberts Truck Products (RTP)

How can managing activities increase efficiency?

How can we determine whether activities are of value to a firm?



QUESTIONS TO THINK ABOUT: Roberts Truck Products (RTP)

What role, if any, do cost reports play in managing activities?



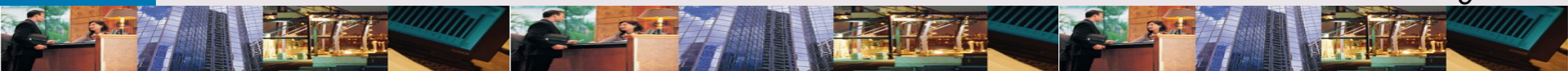
LEARNING OBJECTIVE

1

Describe activity-based management & explain its relationship to activity-based costing.

ACTIVITY-BASED MANAGEMENT: Definition

A systemwide, integrated approach that focuses management's attention on activities for improving customer value and profit.



2-DIMENSIONAL ABM MODEL

2-dimensional model shows intersection of cost & process.

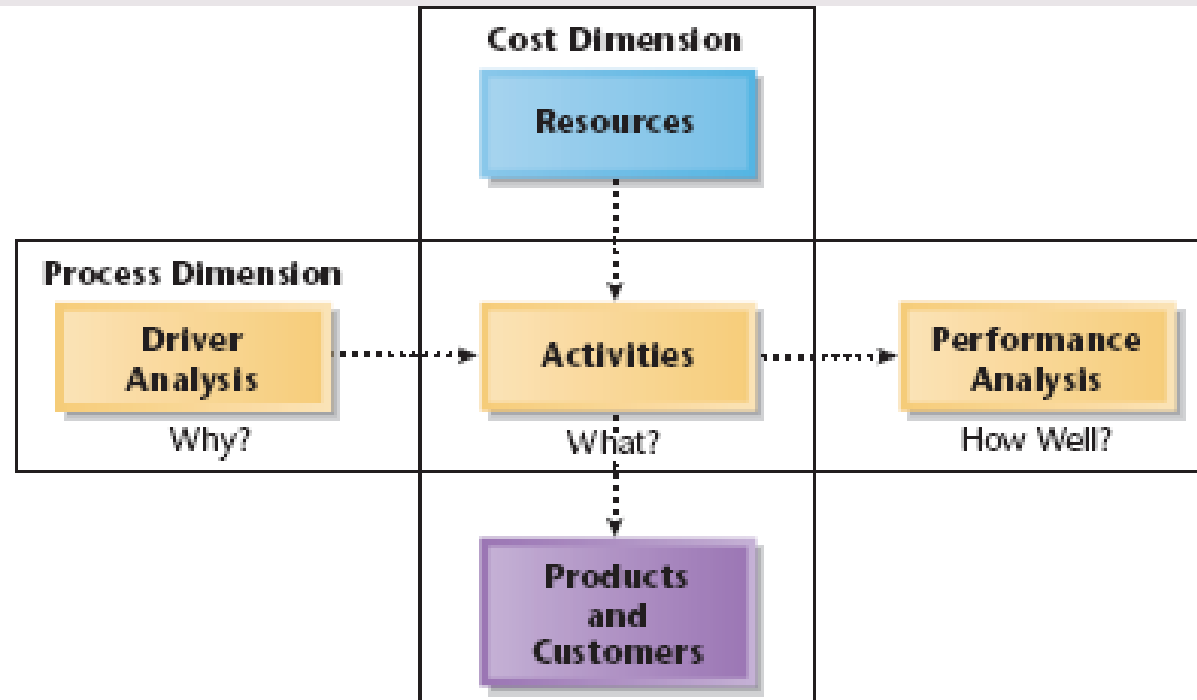
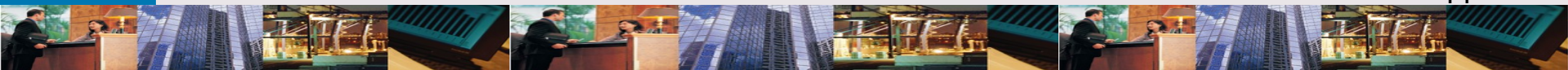


EXHIBIT 5-1

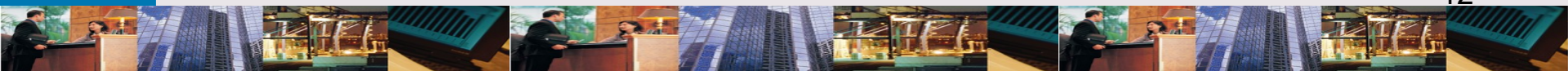
ABM MODEL: Cost Dimension

- ✧ Information about resources, activities, cost objects such as
- ✧ Useful for products, customers, suppliers, distribution channels
- ✧ Objective: improving accuracy of cost assignments



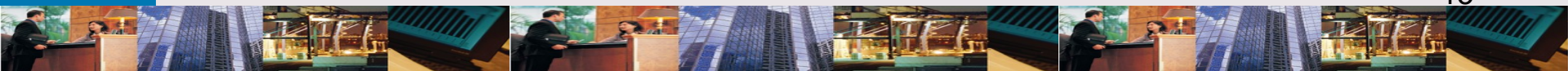
ABM MODEL: Process Dimension

- ✧ Information about
 - ✧ What activities are performed
 - ✧ Why activities are performed
 - ✧ How well they are performed
- ✧ Objective: cost reduction
- ✧ Provides ability to engage in & measure continuous improvement



ABM & ABC

- ✧ ABM incorporates & extends ABC
- ✧ ABM objectives
 - ✧ Improving decision making with accurate cost information
 - ✧ Reducing costs by encouraging, supporting continuous improvements efforts



ABM IMPLEMENTATION MODEL

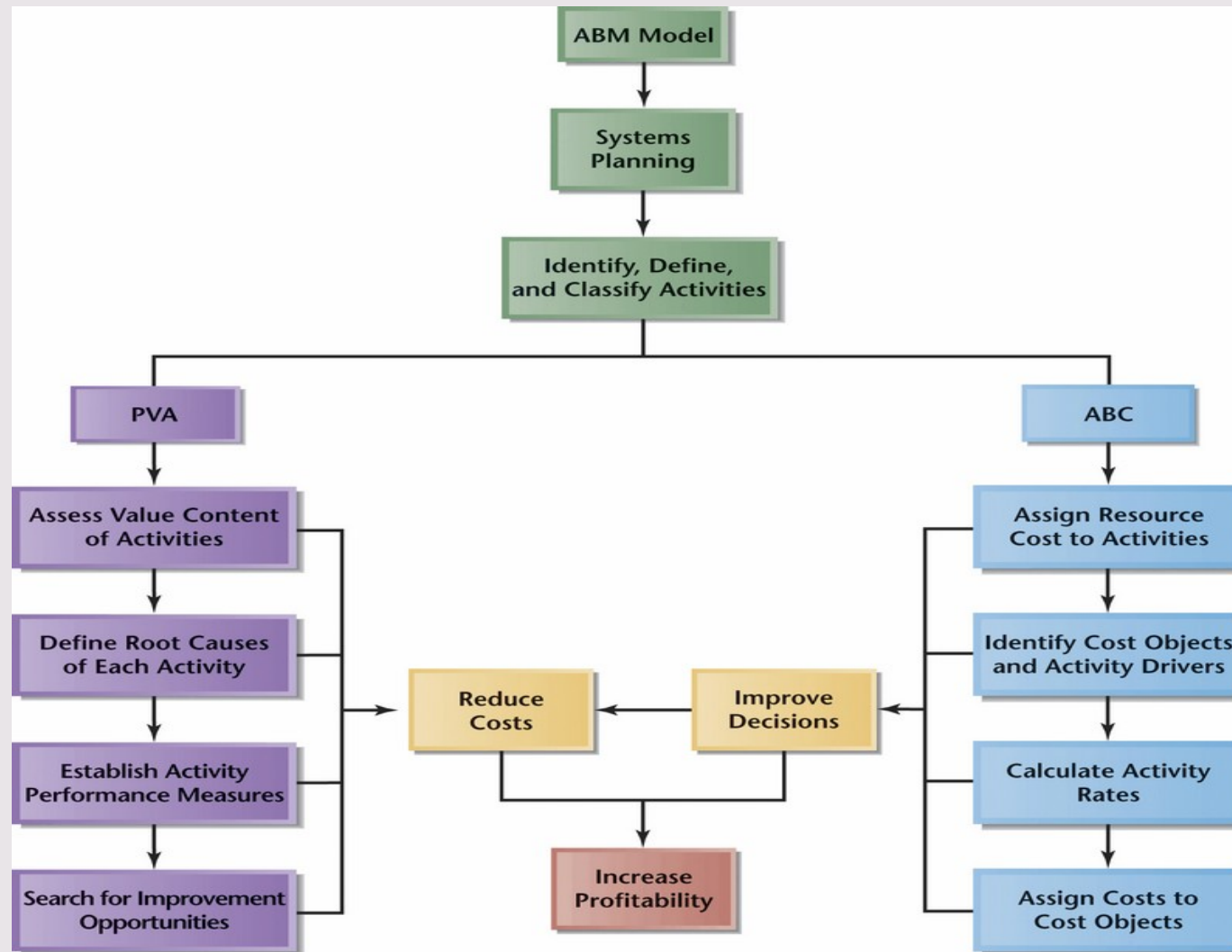
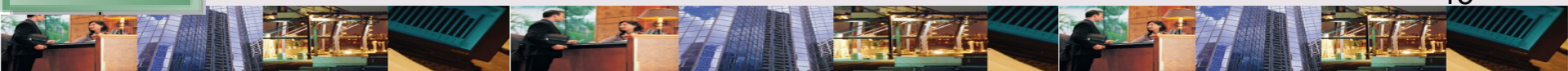


EXHIBIT 5-2

SYSTEMS PLANNING

- ✧ Addresses these issues
 - ✧ Purpose, objectives of ABM system
 - ✧ Organizations current & desired competitive position
 - ✧ Organization's business processes & product mix
 - ✧ Timeline, assigned responsibilities, resources required for implementation
 - ✧ Ability of organization to implement, learn, use new information



ACTIVITY IDENTIFICATION, DEFINITION, CLASSIFICATION

- ✧ Knowing tasks that define activity helps improve efficiency
- ✧ Classifying activities allows ABM to connect with other continuous improvement initiatives
 - ✧ JIT manufacturing
 - ✧ TQM
 - ✧ Total environmental quality cost management

Identify, Define,
and Classify Activities



WHY DOES ABM FAIL?

- ✧ Implementation failure due to
 - ✧ Lack of support from higher level management
 - ✧ Results that do not occur as expected
 - ✧ Significant investment in education, training not made
 - ✧ Resistance to change
 - ✧ Failure to integrate new system



ABM & RESPONSIBILITY ACCOUNTING

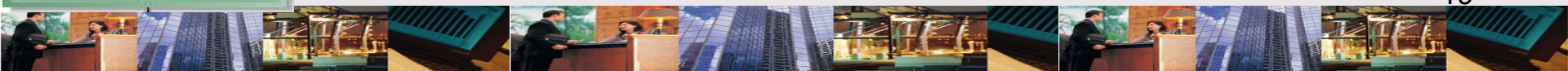
- ✴ Responsibility accounting is fundamental tool of managerial accounting control
- ✴ Also related to process value analysis (PVA)
 - ✴ Assigning responsibility
 - ✴ Establishing performance measures, benchmarks
 - ✴ Evaluating performance
 - ✴ Assigning rewards

Identify, Define, and Classify Activities

RESPONSIBILITY ACCOUNTING SYSTEM

- ✧ 3 ways to assign responsibility and measure performance
 - ✧ Financial (functional)-based system
 - ✧ Activity-based system
 - ✧ Strategic-based system (*Ch. 16*)

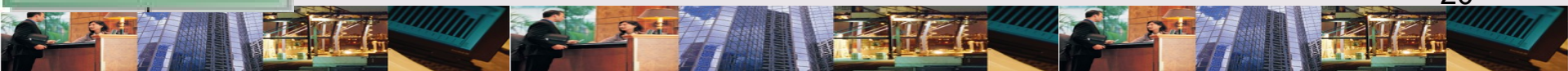
Identify, Define,
and Classify Activities



FINANCIAL-BASED SYSTEMS

- ✧ Assigns responsibilities, measures performance in financial terms
- ✧ Useful in environments with slow or little change
 - ✧ Concrete pipes, blocks
- ✧ Well- defined or relatively stable environments

Identify, Define,
and Classify Activities



ACTIVITY-BASED SYSTEMS

- ✧ Developed for firms in continuous improvement environment
- ✧ Assigns responsibilities to processes
- ✧ Uses both financial & nonfinancial measures of performance
- ✧ Useful in environment that experience rapid change
- ✧ Computer technology

Identify, Define,
and Classify Activities



COMPARING SYSTEMS:

Assigning Responsibilities

Assignment
emphasizes
improvement

Financial-Based Responsibility	Activity-Based Responsibility
1. Organizational units 2. Local operating efficiency 3. Individual accountability 4. Financial outcomes	1. Processes 2. Systemwide efficiency 3. Team accountability 4. Financial outcomes

Exhibit 5-4 Responsibility Assignments Compared

Assignment based
on function

EXHIBIT 5-4

Identify, Define,
and Classify Activities

COMPARING SYSTEMS:

Measuring Performance

Process
improvement
measures

Financial-Based Measures

1. Organizational unit budgets
2. Standard costing
3. Static standards
4. Currently attainable standards

Activity-Based Measures

1. Process-oriented standards
2. Value-added standards
3. Dynamic standards
4. Optimal standards

Exhibit 5-5 Performance Measures Compared

Financial
performance
measures

EXHIBIT 5-5

Identify, Define,
and Classify Activities

COMPARING SYSTEMS:

Evaluating Performance

Evaluates time, quality, efficiency & financial standards

Financial-Based Performance Evaluation	Activity-Based Performance Evaluation
1. Financial efficiency 2. Controllable costs 3. Actual versus standard 4. Financial measures	1. Time reductions 2. Quality improvements 3. Cost reductions 4. Trend measurement

Exhibit 5-6 Performance Evaluation Compared

Meet/beat financial standard

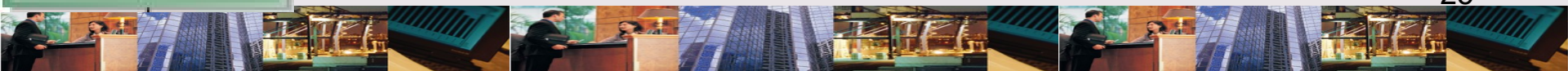
EXHIBIT 5-6

Identify, Define, and Classify Activities

GAINSHARING: Definition

ABM system allows employees to share in gains related to specific improvement projects.

Identify, Define,
and Classify Activities



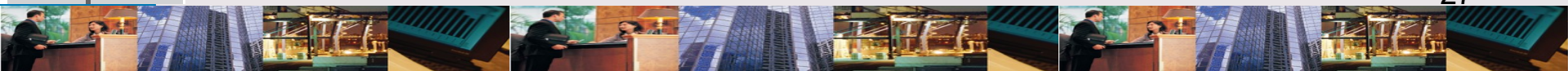
LEARNING OBJECTIVE

2

Explain process value analysis.

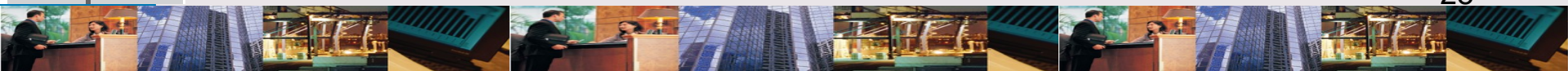
PROCESS VALUE ANALYSIS: Definition

PVA emphasizes accountability
for activities rather than costs;
focuses on systemwide
performance.



PVA CONCERNS

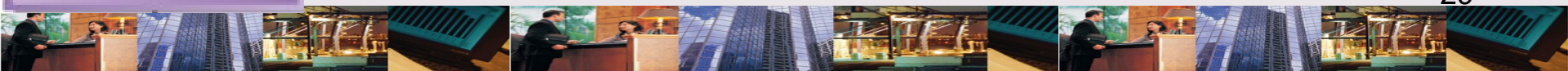
- ✧ Driver analysis
- ✧ Activity analysis
- ✧ Activity performance measurement



ROOT CAUSES: Definition

Understanding what causes activity costs by understanding activity inputs & outputs; most basic causes for an activity being performed.

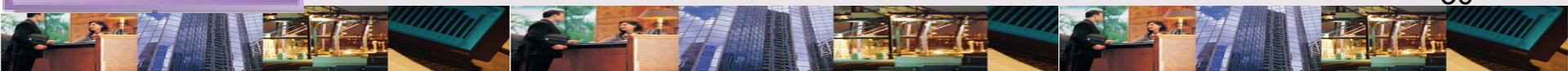
Define Root Causes
of Each Activity



Why are “root causes” so important?

Because the root cause of 1 activity may be root cause of related activities.

Define Root Causes
of Each Activity



ACTIVITY ANALYSIS: Definition

The process of identifying, describing, evaluating the activities an organization performs.

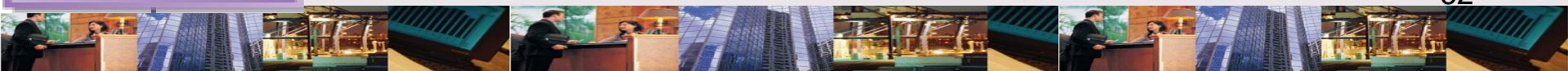
Assess Value Content
of Activities



ACTIVITY ANALYSIS OUTCOMES

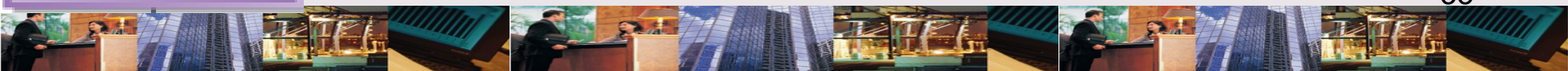
- ✧ What activities are done
- ✧ How many people perform the activities
- ✧ Time, resources required to perform activities
- ✧ Assessing value of activities to organization
- ✧ Activities can be classified as
 - ✧ Value-added
 - ✧ Non-value-added

Assess Value Content
of Activities



What are “value-added” activities?

Value-added activities are activities that are necessary to remain in business.



VALUE-ADDED ACTIVITIES

✧ Are

✧ Mandatory to comply with laws

✧ Discretionary

✧ Produces a change of state

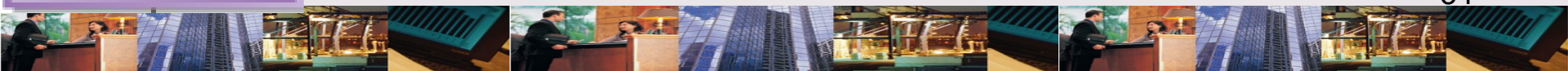
✧ Not achievable by preceding activities

✧ Enables other activities to be performed

✧ Performed at a value-added cost to achieve perfect efficiency

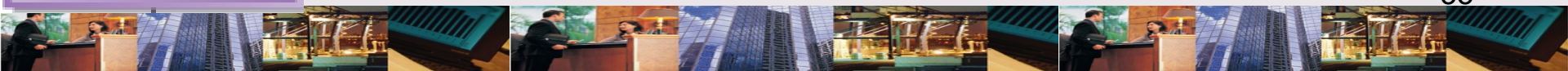
✧ Eliminate waste & reduce costs

Assess Value Content
of Activities



NON-VALUE-ADDED ACTIVITIES

- ✧ Are **un**necessary
- ✧ Fail to satisfy 3 defining conditions of value-added activities
- ✧ Incur non-value-added costs of inefficiency

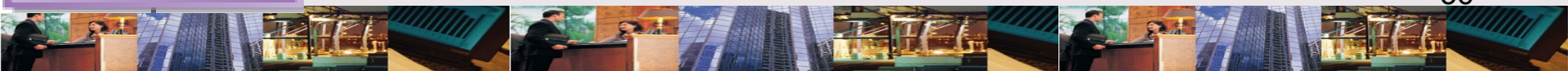


NON-VALUE-ADDED ACTIVITIES:

Examples

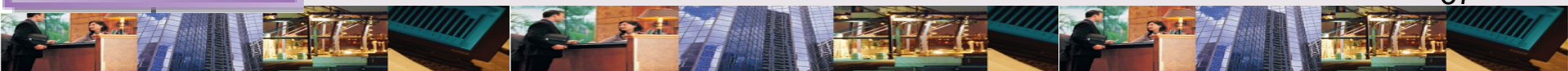
- ✧ Scheduling
- ✧ Moving
- ✧ Waiting
- ✧ Inspecting
- ✧ Storing

Challenge of activity analysis: produce goods without using non-value-added activities.



KAIZEN COSTING: Definition

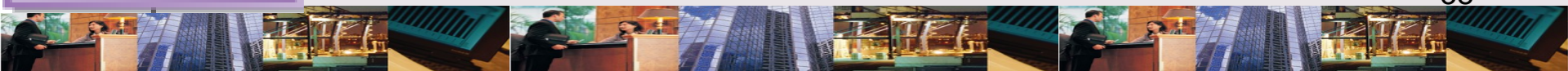
The effort to reduce costs of existing products & processes.



REDUCING COSTS

- ✧ Activity elimination
 - ✧ Focusing on non-value-added activities
- ✧ Activity selection
 - ✧ Choosing among different sets of activities
- ✧ Activity reduction
 - ✧ Reducing time, resources required
- ✧ Activity sharing
 - ✧ Using economies of scale

Assess Value Content
of Activities



ACTIVITY PERFORMANCE MEASUREMENT: Definition

Assessing how well an activity was performed and results achieved using both financial & nonfinancial measures.

ACTIVITY PERFORMANCE MEASURES

✧ Efficiency

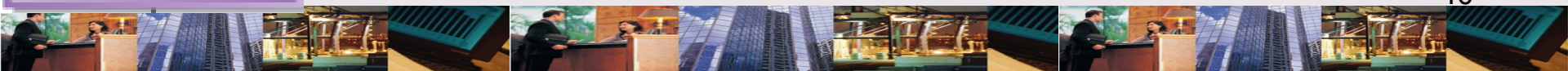
✧ Relationship of activity inputs & outputs

✧ Quality

✧ Doing it right the first time

✧ Time

✧ Shortening activity time



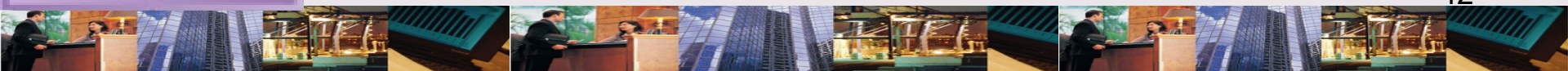
LEARNING OBJECTIVE

3

Describe activity
performance
measurement.

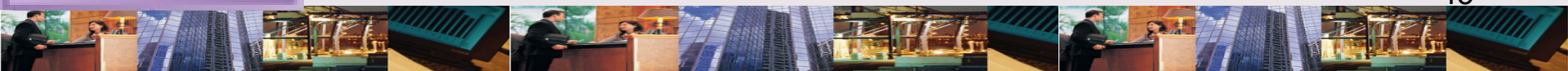
FINANCIAL MEASURES OF ACTIVITY PERFORMANCE

- ✧ For potential & actual savings
 - ✧ Value- & non-value-added activity cost reports
 - ✧ Trends in activity cost reports
 - ✧ Kaizen standard setting
 - ✧ Benchmarking
 - ✧ Life-cycle costing



VALUE-ADDED STANDARD

- ✧ Calls for elimination of non-value-added activities
- ✧ Identifies optimal activity output
- ✧ Compares actual to value-added activity costs allowing management to
 - ✧ Assess level of activity inefficiency
 - ✧ Determine potential for improvement



FORMULAS

SQ: value-added output level

SP: standard price per output measure

AQ: actual quantity used of flexible resources

$$\text{Value-added} = \text{SQ} \times \text{SP}$$

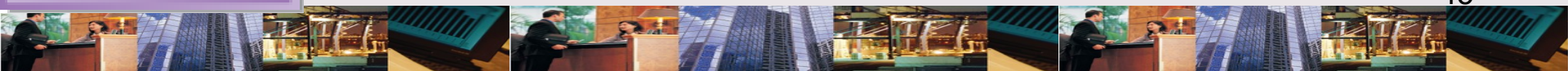
$$\text{Non-value-added costs} = (\text{AQ} - \text{SQ})\text{SP}$$

EXHIBIT 5-8

COST REPORT: Step 1

Activity	Activity Driver	SQ*	AQ*	SP*
Welding	Welding hours	10,000	12,000	\$40
Rework	Rework hours	0	10,000	9
Setups	Setup hours	0	6,000	60
Inspection	# Inspections	0	4,000	15

Search for Improvement
Opportunities



COST REPORT: Step 2

Activity	Value-Added Costs	Non-Value-Added Costs	Actual Costs
Welding	\$400,000	\$80,000	\$480,000
Rework	0	90,000	90,000
Setups	0	360,000	360,000
Inspection	<u>0</u>	<u>60,000</u>	<u>60,000</u>
Total	<u>\$ 400,000</u>	<u>\$ 590,000</u>	<u>\$ 990,000</u>

EXHIBIT 5-9

Search for Improvement Opportunities

COST REPORT: Step 2

Activity	Value-Added	Non-Value-Added	Actual Costs
Welding	\$400,000	\$80,000	\$480,000
Rework	0	90,000	90,000
Setups	0	360,000	360,000
Inspection	<u>0</u>	<u>60,000</u>	<u>60,000</u>
Total	<u>\$ 400,000</u>	<u>\$ 590,000</u>	<u>\$ 990,000</u>

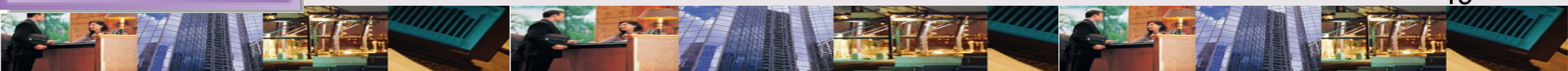
Cost report emphasizes the opportunity for improvement.

EXHIBIT 5-9

Search for Improvement Opportunities

TREND REPORTING

Allows management to follow up on actions taken to reduce costs by examining whether outcomes were as expected.



TREND REPORT: Step 3

Non-Value-Added Costs

Activity	Last Year	Current Year	Change
Welding	\$80,000	\$50,000	\$30,000
Rework	90,000	70,000	20,000
Setups	360,000	200,000	160,000
Inspection	<u>60,000</u>	<u>35,000</u>	<u>25,000</u>
Total	\$ <u>590,000</u>	\$ <u>355,000</u>	\$ <u>235,000</u>

Trend report shows improvement that has been made.

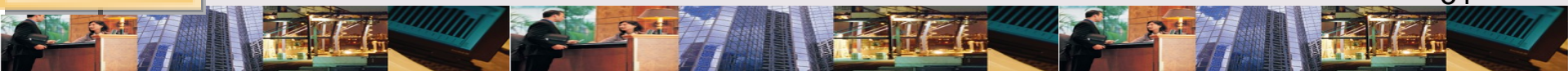
EXHIBIT 5-10

Search for Improvement Opportunities

What is kaizen costing?

Kaizen costing helps reduce costs by repeated use of 2 subcycles: 1) continuous improvement, and 2) maintenance.

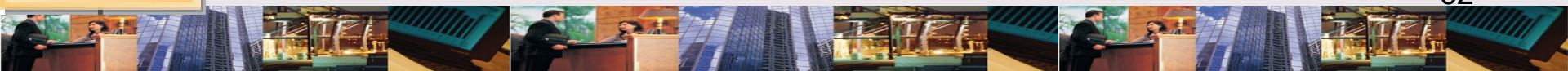
Reduce
Costs



BENCHMARKING: Definition

Uses “best practices” as the standard for evaluating activity performance with the goal of becoming the best at performing activities & processes.

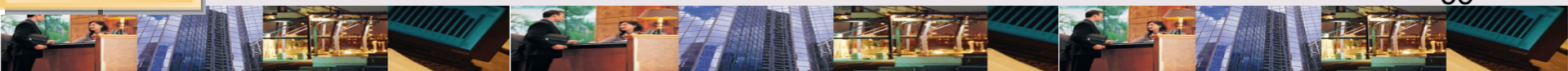
Reduce
Costs



ACTIVITY CAPACITY : Definition

Activity drivers measure activity capacity, that is the number of times an activity can be performed.

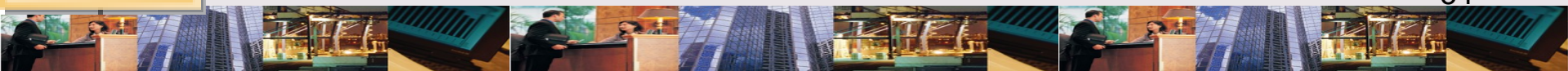
Reduce
Costs



CAPACITY VARIANCES

Unused capacity, the difference between activity availability & activity usage, needs management attention to reduce costs.

Reduce
Costs



LEARNING OBJECTIVE

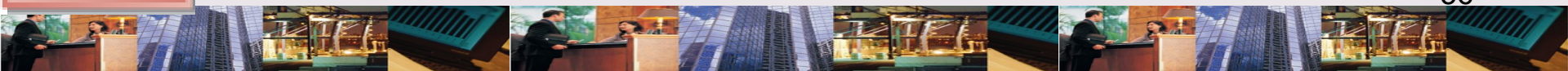
4

Describe activity-based customer & supplier costing.

ACTIVITY-BASED CUSTOMER & SUPPLIER COSTING

By applying the approach of activity-based costing to customers and suppliers, managers can identify & reduce true cost of these relationships.

Increase
Profitability



ACTIVITY-BASED CUSTOMER COSTING: An Example

	Large Customer	10 Smaller Customers
Order-filling costs	\$ 4,000	\$ 400,000
Sales force costs	10,000	210,000

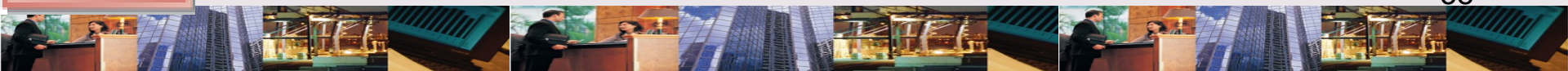
ABC costing shows comparative cost of larger & smaller customers, assuming same number of units sold.

Increase
Profitability

ACTIVITY-BASED SUPPLIER COSTING

- ✧ Identifies costs other than price such as
 - ✧ Quality
 - ✧ Reliability
 - ✧ Delivery timeliness
- for management to consider when selecting suppliers

Increase
Profitability



CHAPTER 5

THE END